Design for Testability and Scan Design

Nguyen-Van Thanh-Loc

Email: locnvt@hcmute.edu.vn

Outline

- Design for Testability (DFT)
- Test Point Insertion (TPI)
- Structured DFT Approach
- Scan Design Operating Modes
- Testing Difficulty in Sequential Circuits
- Scan Design Concept
- Scan Cell Designs
- Full-Scan Design
- Full-Scan Test Operation

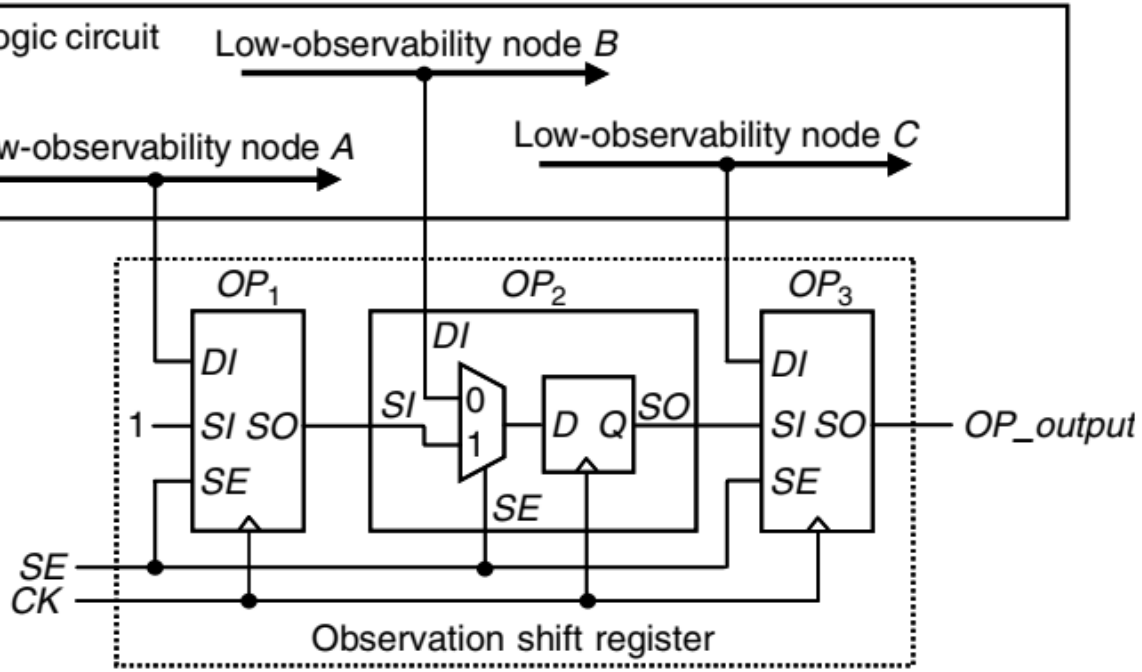
Design for Testability (DFT)

- Because many internal states exist, setting a sequential circuit to a required internal state can require a very large number of input events.
- The structured approach for testability improvement was introduced to allow DFT engineers to follow a methodical process for improving the testability of a design.
- Scan design has been found to be one of the most effective structured DFT methodologies for testability improvement.

Test Point Insertion (TPI) (1/3)

- Test point insertion (TPI) is a commonly used ad hoc DFT technique for improving the controllability and observability of internal nodes.
- Testability analysis is typically used to identify the internal nodes where test points should be inserted, in the form of control or observation points.
- As a result, the controllability and observability of the circuit nodes are dramatically improved.

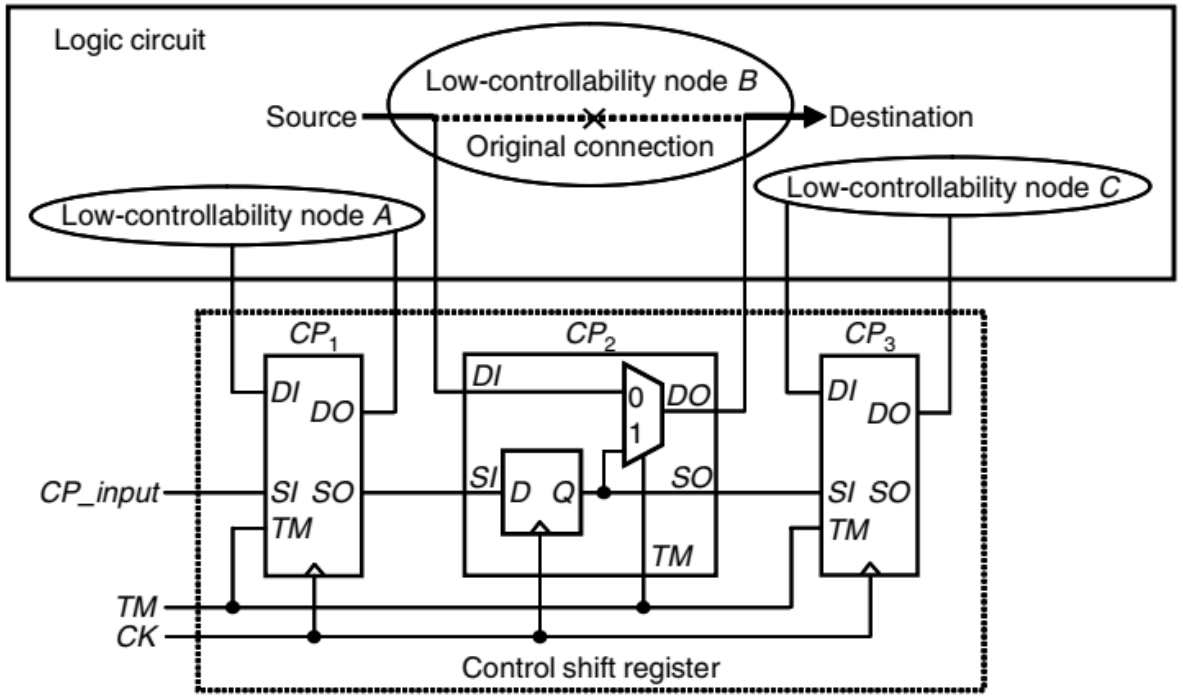
Test Point Insertion (TPI) (2/3)



■ **FIGURE 2.5**

Observation point insertion.

Test Point Insertion (TPI) (3/3)



■ **FIGURE 2.6**

Control point insertion.

Structured DFT Approach

- The structured DFT approach attempts to improve the overall testability of a circuit with a test-oriented design methodology.
- Scan design, the most widely used structured DFT methodology, attempts to improve testability of a circuit by improving the controllability and observability of storage elements in a sequential design.
- Typically, this is accomplished by converting the sequential design into a scan design with three modes of operation: normal mode, shift mode, and capture mode.

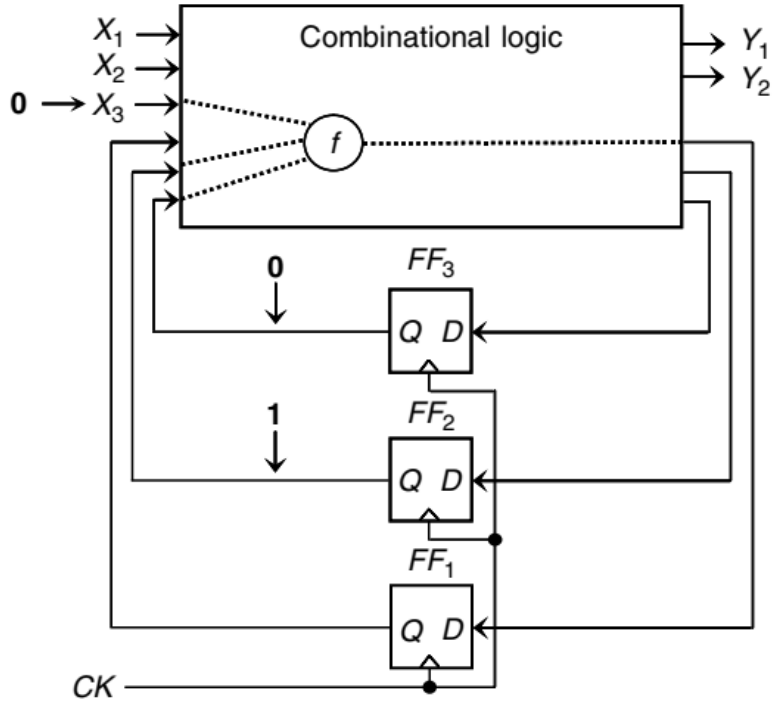
Scan Design Operating Modes

- In normal mode, all test signals are turned off, and the scan design operates in the functional configuration.
- In both shift and capture modes, a test mode signal (TM) is often used to turn on all test-related fixes that are necessary to simplify the test, debug, and diagnosis tasks.
- These circuit modes and operations are distinguished using additional test signals or test clocks.

Testing Difficulty in Sequential Circuits

- Assume that a stuck-at fault f in the combinational logic requires the primary input X3, flip-flop FF2, and flip-flop FF3 to be set to 0, 1, and 0, respectively, to capture the fault effect into FF1.
- Because the values stored in FF2 and FF3 are not directly controllable from the primary inputs, a long sequence of operations may have to be applied in order to set FF2 and FF3 to the required values.
- Furthermore, in order to observe the fault effect on the captured value in flip-flop FF1, a long checking experiment may be required to propagate the value of FF1 to a primary output.

Testing Difficulty in Sequential Circuits



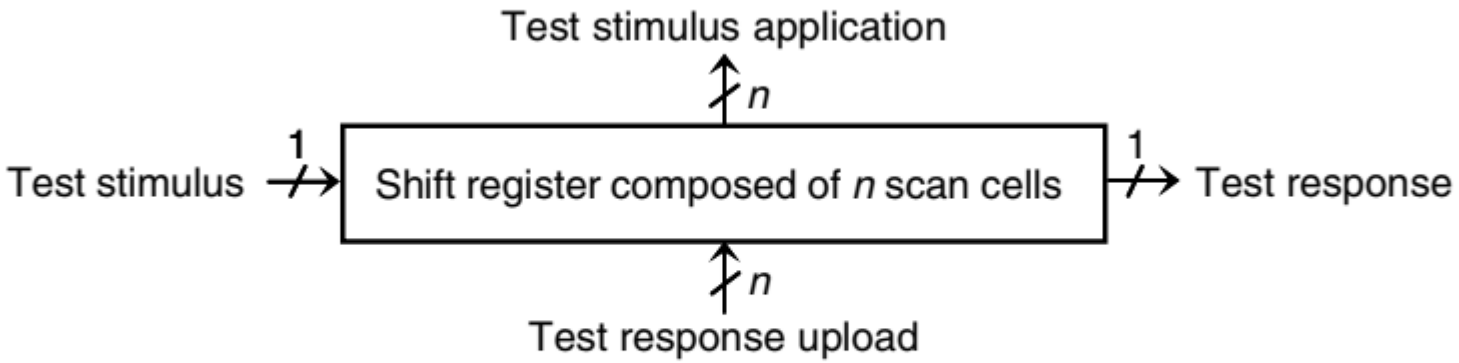
■ **FIGURE 2.7**

Difficulty in testing a sequential circuit.

Scan Design Concept

- Scan design attempts to ease this difficulty by providing external access to selected storage elements in a design.
- This is accomplished by first converting selected storage elements in the design into scan cells and then stitching them together to form one or more shift registers, called scan chains.
- In the scan design illustrated in Figure 2.8, the n storage elements are now configured as a shift register in shift mode.

Scan Design Concept



■ **FIGURE 2.8**

Scan design concept.

Scan Cell Designs

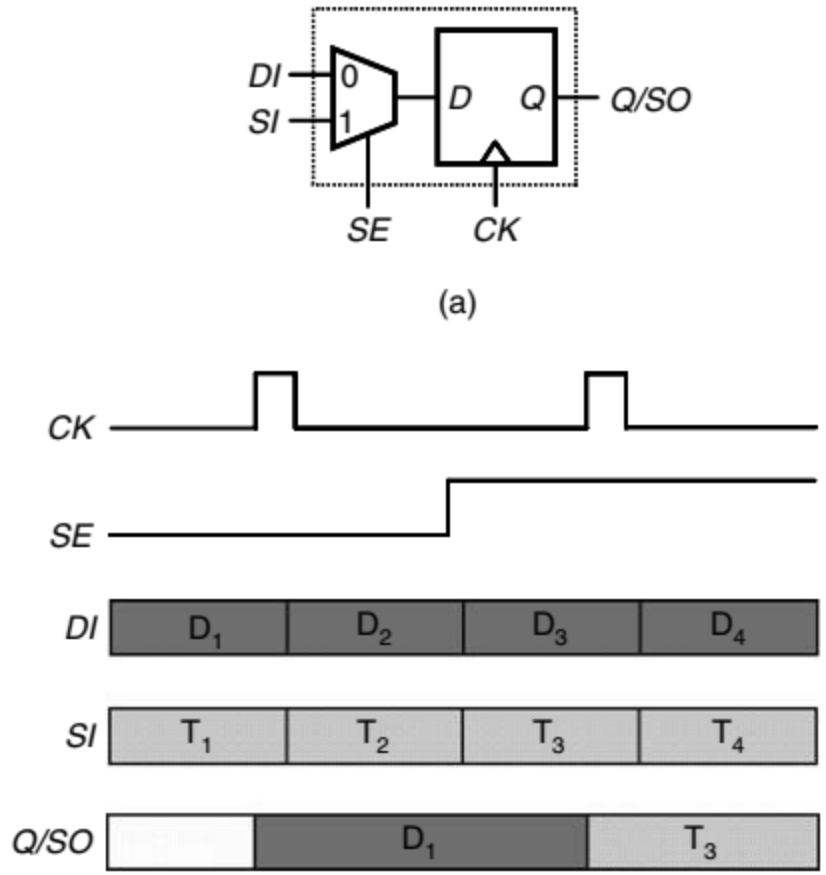
- A scan cell has two different input sources: a data input driven by the combinational logic and a scan input driven by another scan cell.
- A selection mechanism must be provided to allow a scan cell to operate in two different modes: normal/capture mode and shift mode.
- In this section, we describe three widely used scan cell designs: muxed-D scan, clocked-scan, and level-sensitive scan design (LSSD).

Muxed-D Scan Cell

- The most widely used scan cell replacement for the D storage element is the muxed-D scan cell.
- Figure 2.9a shows an edge-triggered muxed-D scan cell design.
- This scan cell is composed of a D flip-flop and a multiplexer. The multiplexer uses a scan enable (SE) input to select between the data input (DI) and the scan input (SI).

Muxed-D Scan Cell

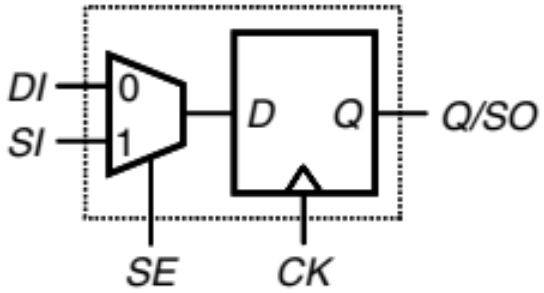
- In normal/capture mode, SE is set to 0.
- The value present at the data input DI is captured into the internal D flip-flop when a rising clock edge is applied.
- In shift mode, SE is set to 1. The SI is now used to shift in new data to the D flip-flop while the content of the D flip-flop is being shifted out.



■ **FIGURE 2.9**

Edge-triggered muxed-D scan cell design and operation: (a) edge-triggered muxed-D scan cell, and (b) sample waveforms.

Design Muxed-D Scan Cell by Verilog (1/4)



Scan Cell

Truth table

SE	↑CK	DI	SI	Q(next)	Mode
0	↑	0	X	0	Normal/Capture
0	↑	1	X	1	Normal/Capture
1	↑	X	0	0	Shift
1	↑	X	1	1	Shift
X	No edge	X	X	Q(previous)	Hold

Design Muxed-D Scan Cell by Verilog (1/4)

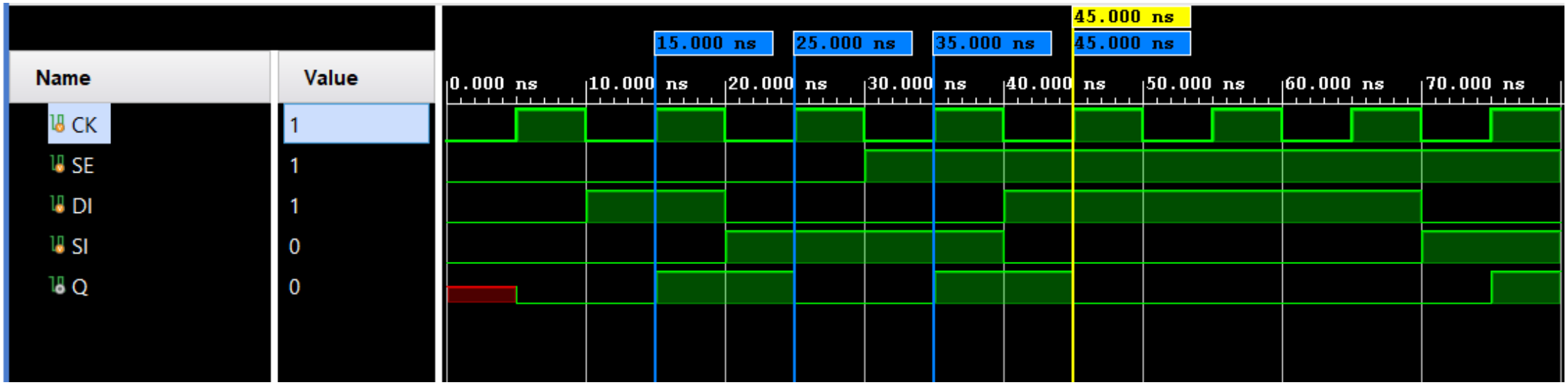
```
module D_scan_cell(
  input wire CK,
  input wire SE,
  input wire DI,
  input wire SI,
  output reg Q );
  always @(posedge CK) begin
      if(SE) Q <= SI; // Shift Mode
      else Q <= DI; // Normal/Capture Mode
  end
endmodule
```

Design Muxed-D Scan Cell by Verilog (1/4)

```
module tb_D_scan_cell();
reg CK; reg SE;
reg DI; reg SI;
wire Q;
D_scan_cell uut( .CK(CK), .SE(SE), .DI(DI),
.SI(SI), .Q(Q) );
// Clock generation always #5 CK = ~CK;
initial
begin
CK = 0;
SE = 0;
DI = 0;
SI = 0;
#10;
```

```
//------------------------------------ //
Normal Mode //-----------------------------
---------
DI = 1; SI = 0; #10;
DI = 0; SI = 1; #10;
//------------------------------------ //
Shift Mode //-----------------------------
--------
SE = 1; SI = 1; DI = 0; #10;
SI = 0; DI = 1; #10;
//------------------------------------ //
Back to Normal //-------------------------
------------
DI = 1; SI = 0; #10;
DI = 0; SI = 1; #10;
$finish;
end endmodule
```

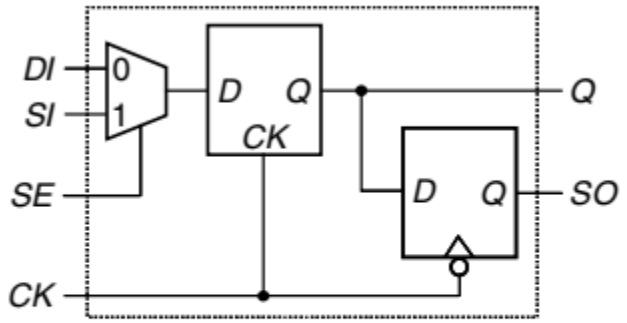
Design Muxed-D Scan Cell by Verilog (4/4)



Level-Sensitive / Edge-Triggered Muxed-D Scan Cell

- Figure 2.10 shows a level-sensitive/edge-triggered muxed-D scan cell design, which can be used to replace a D latch in a scan design.
- This scan cell is composed of a multiplexer, a D latch, and a D flip-flop.
- **Shift** operation is conducted in an **edge**-triggered manner, while **normal** operation and capture operation are conducted in a **level-sensitive** manner.

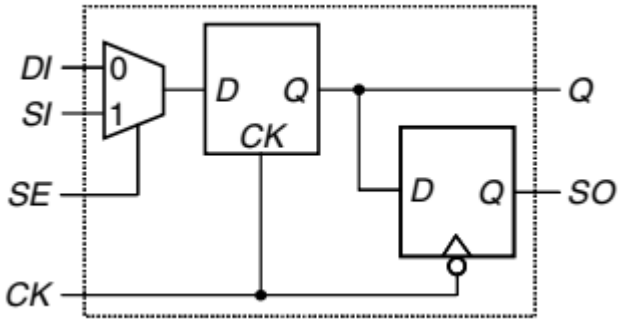
Level-Sensitive / Edge-Triggered Muxed-D Scan Cell



■ **FIGURE 2.10**

Level-sensitive/edge-triggered muxed-D scan cell design.

Level-Sensitive / Edge-Triggered Muxed-D Scan Cell



Scan Cell

Truth table

SE	CK	DI	SI	Q	SO
0	0	X	X	Hold	Hold
0	1	0	X	0	Hold
0	1	1	X	1	Hold
1	↑	X	0	Hold	0
1	↑	X	1	Hold	1

Level-Sensitive / Edge-Triggered Muxed-D Scan Cell

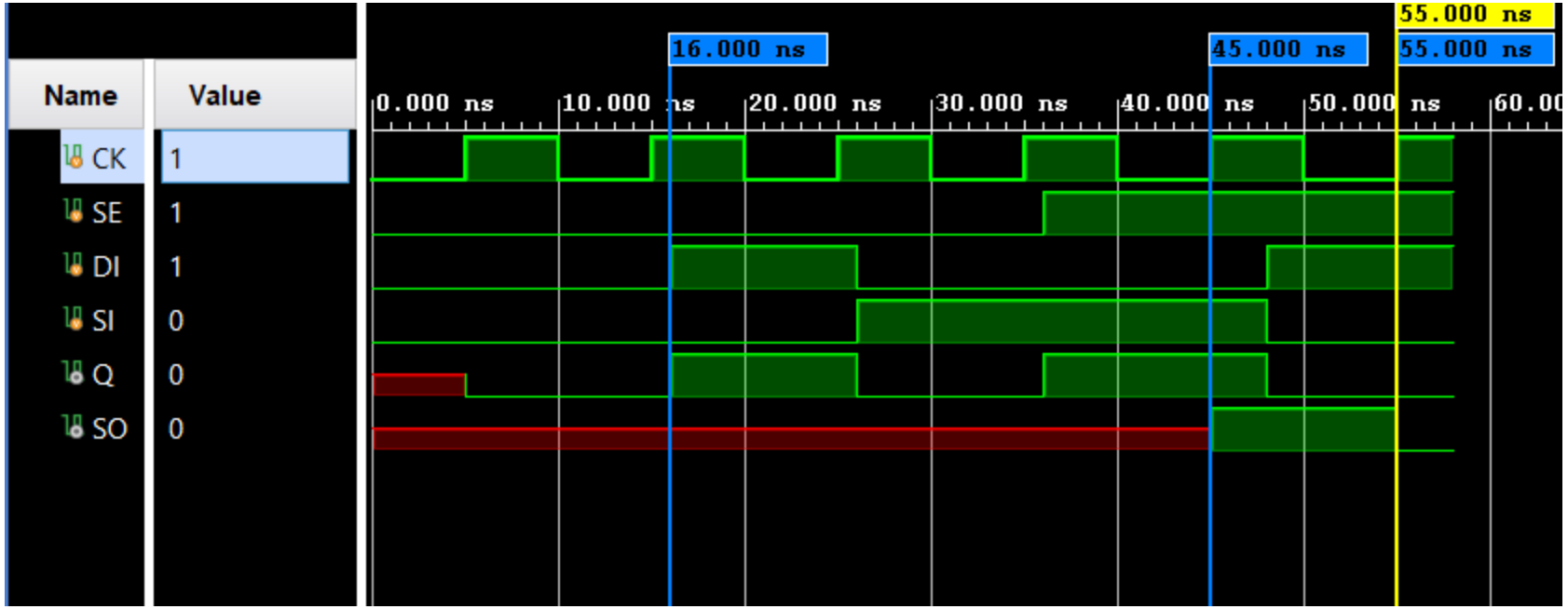
```
module ls_edge_scan_cell( input CK, input SE, input DI, input SI, output reg Q, output reg SO );
//---------------------------- // Level-sensitive latch //----------------------------
always @(CK or SE or DI) begin
if(CK) begin
        if(SE) Q <= SI;
else Q <= DI;
end
end
//---------------------------- // Edge-triggered scan FF //----------------------------
always @(posedge CK) begin
        if(SE) SO <= Q;
end
endmodule
```

Level-Sensitive / Edge-Triggered Muxed-D Scan Cell

```
module tb();
reg CK;
reg SE;
reg DI;
reg SI;
wire Q;
wire SO;
ls_edge_scan_cell uut( .CK(CK), .SE(SE),
.DI(DI), .SI(SI), .Q(Q), .SO(SO) );
always #5 CK=~CK;
initial
begin CK=0; SE=0; DI=0; SI=0; #16;
```

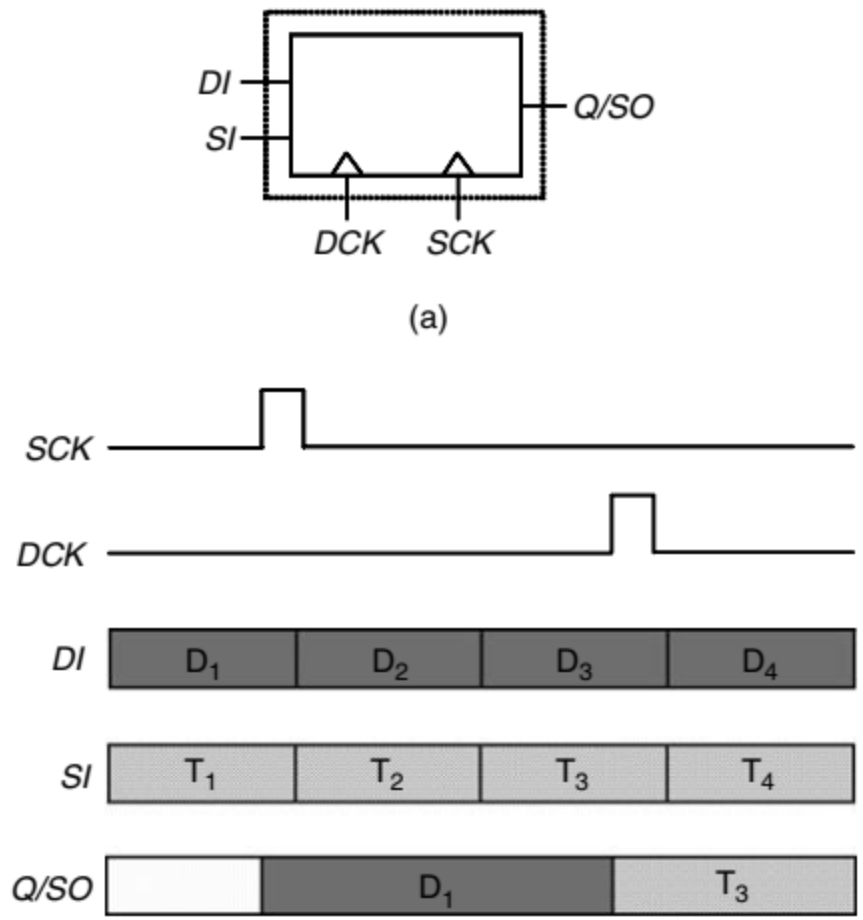
```
////////////////////////// // Normal Mode
//////////////////////////
DI=1;SI=0; #10; DI=0;SI=1; #10;
////////////////////////// // Shift Mode
//////////////////////////
SE=1;#2; SI=1;DI=0;#10; SI=0;DI=1;#10;
$finish;
end
endmodule
```

Level-Sensitive / Edge-Triggered Muxed-D Scan Cell



Clocked-Scan Cell

- In the clocked-scan cell, input selection is conducted using two independent clocks, data clock DCK and shift clock SCK, as shown in Figure 2.11a.
- In normal/capture mode, the data clock DCK is used to capture the value present at the data input DI into the clocked-scan cell.
- In shift mode, the shift clock SCK is used to shift in new data from the scan input SI into the clocked-scan cell, while the current content of the clocked-scan cell is being shifted out.

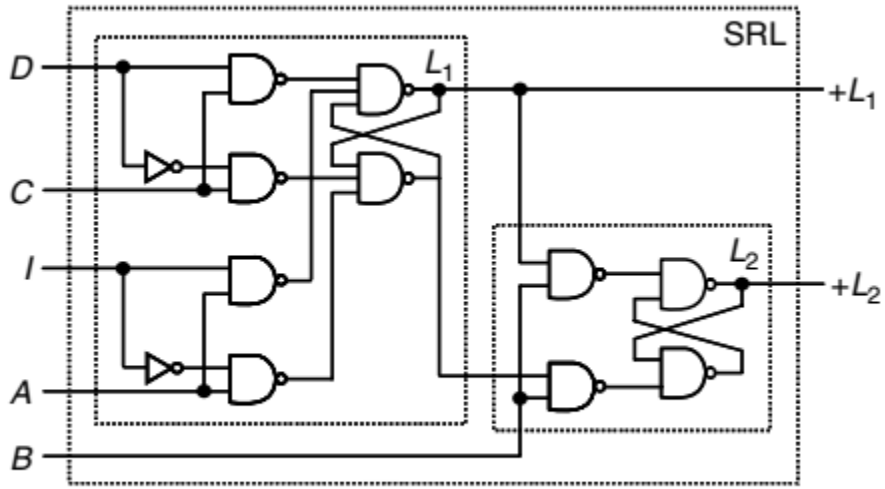


■ **FIGURE 2.11**

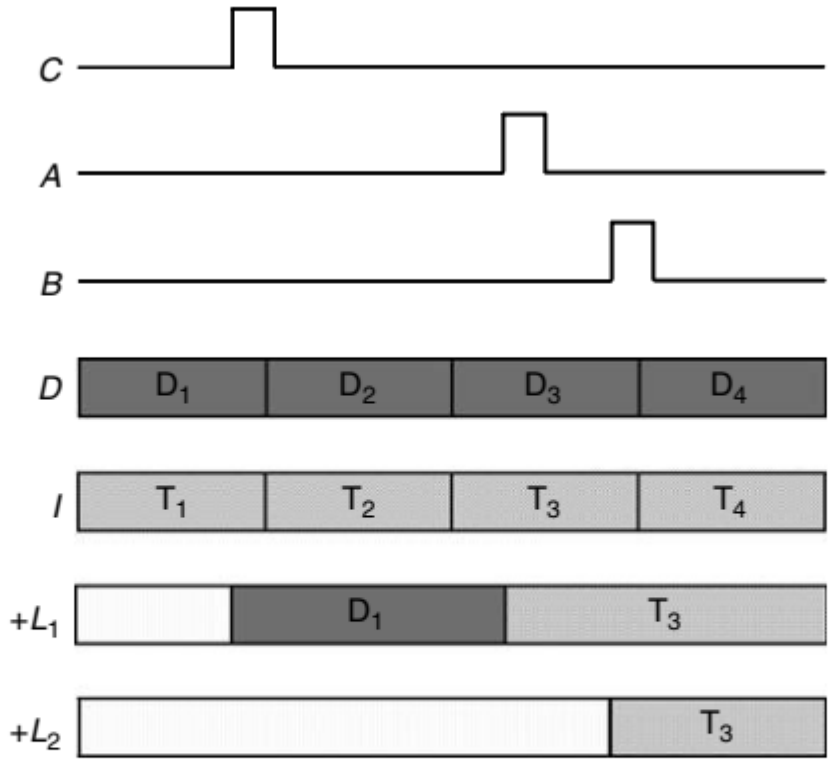
Clocked-scan cell design and operation: (a) clocked-scan cell, and (b) sample waveforms.

LSSD Scan Cell

- Figure 2.12a shows a polarity-hold shift register latch (SRL) design that can be used as an LSSD scan cell. This scan cell contains two latches, a master two-port D latch L1 and a slave D latch L2.
- In order to guarantee race-free operation, clocks A, B, and C are applied in a nonoverlapping manner.
- In shift mode, clocks A and B are used to latch scan data from the scan input I and to output this data onto +L2, which is then used to drive the scan input of the next scan cell.



(a)



(b)

■ **FIGURE 2.12**

Polarity-hold SRL design and operation: (a) polarity-hold SRL, and (b) sample waveforms.

Scan Architectures

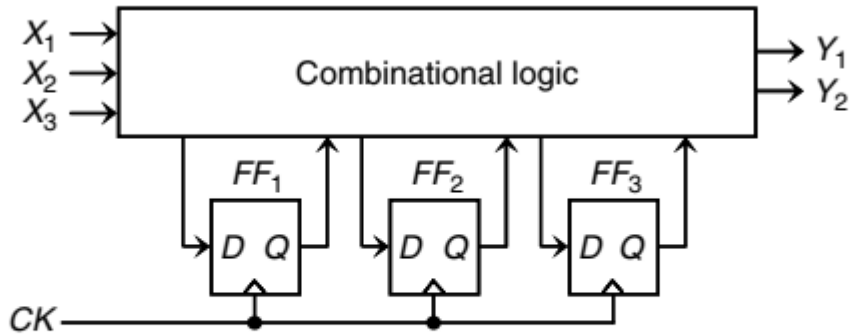
- Three popular scan architectures are full-scan design, partial-scan design, and random-access scan design.
- In full-scan design, all storage elements are converted into scan cells and combinational ATPG is used for test generation.
- In partial-scan design, a subset of storage elements is converted into scan cells, while in random-access scan design, a random addressing mechanism is used to provide direct access to read or write any scan cell.

Full-Scan Design

- In full-scan design, all storage elements are replaced with scan cells, which are then configured as one or more shift registers (scan chains) during the shift operation.
- As a result, all inputs to the combinational logic can be controlled and all outputs from the combinational logic can be observed.
- The main advantage of full-scan design is that it converts the difficult problem of sequential ATPG into the simpler problem of combinational ATPG.

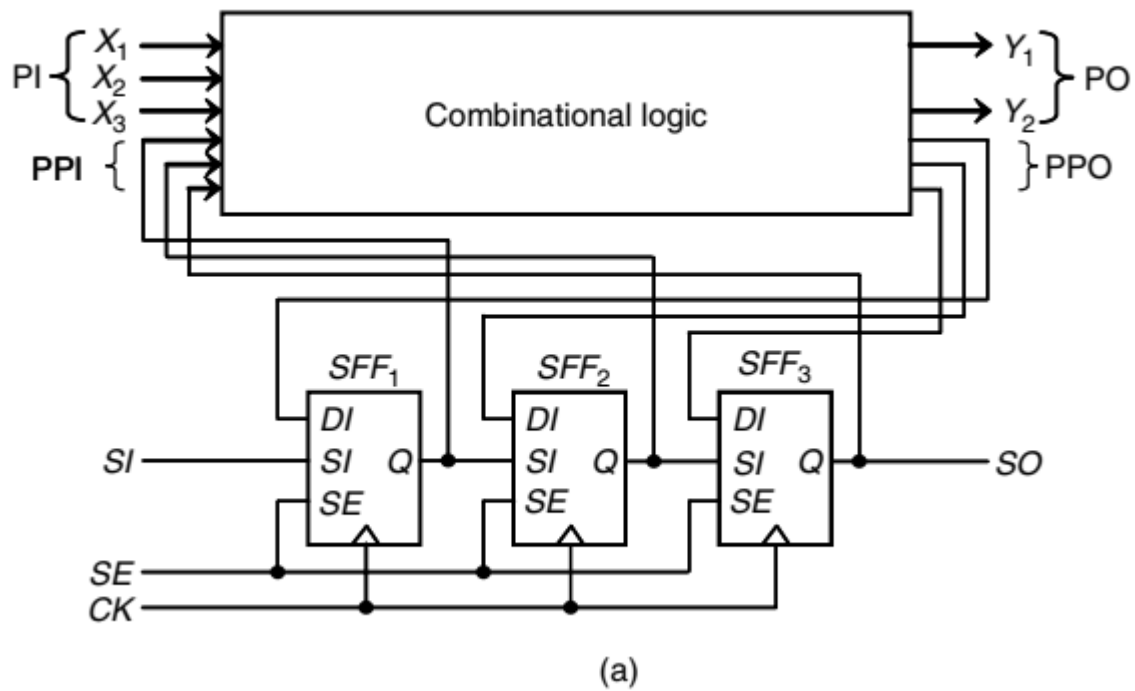
Muxed-D Full-Scan Design

- The three D flip-flops, FF1, FF2, and FF3, are replaced with three muxed-D scan cells, SFF1, SFF2, and SFF3, respectively.
- To form a scan chain, the scan inputs SI of SFF2 and SFF3 are connected to the outputs Q of the previous scan cells, while the first scan cell is connected to SI and the last scan cell is connected to SO.
- In shift mode, SE is set to 1 and the scan cells operate as a single scan chain, whereas in capture mode, SE is set to 0 and the scan cells capture the test response from the combinational logic.

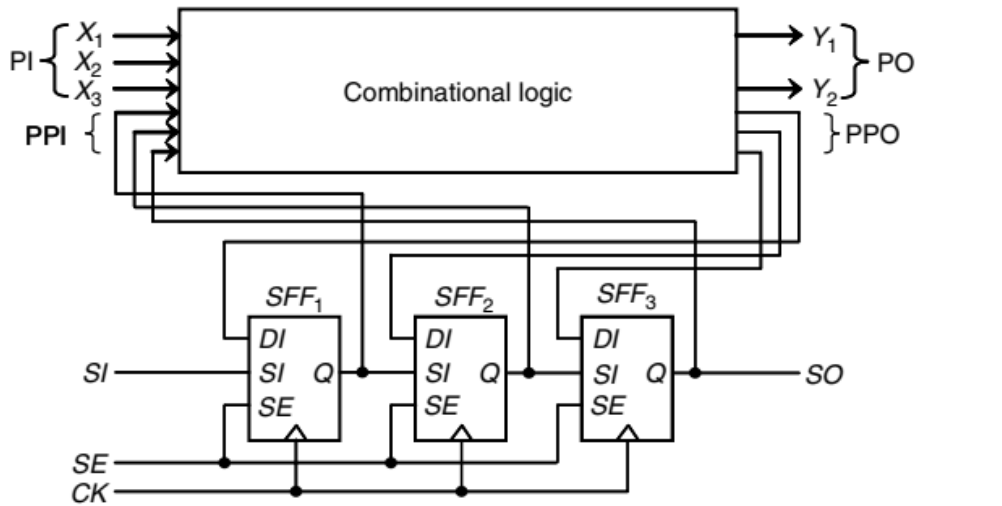


■ **FIGURE 2.13**

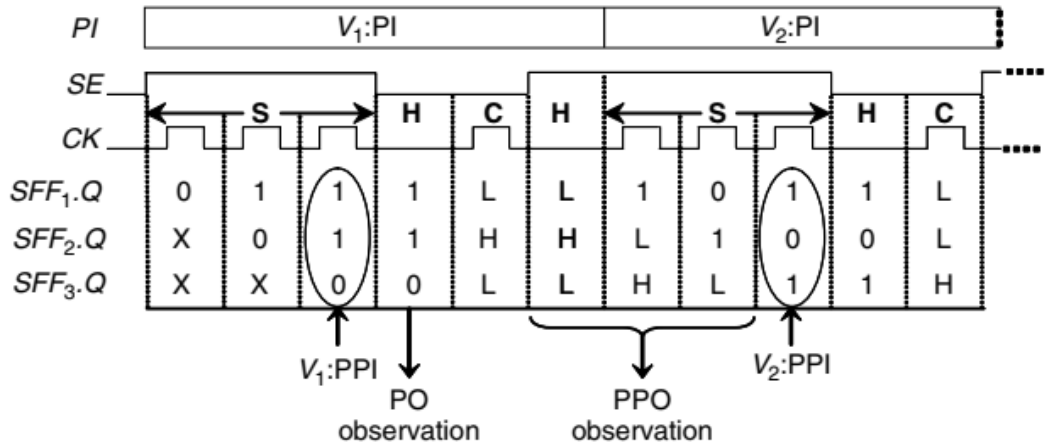
Sequential circuit example.



(a)



(a)



S: shift operation / **C**: capture operation / **H**: hold cycle

(b)

■ **FIGURE 2.14**

Muxed-D full-scan circuit and test operations: (a) muxed-D full-scan circuit, and (b) test operations.

PI, PPI, PO, and PPO in Full-Scan Design

- The combinational logic in a full-scan circuit has two types of inputs: primary inputs (PIs) and pseudo primary inputs (PPIs).
- Primary inputs refer to the external inputs to the circuit, while pseudo primary inputs refer to the scan cell outputs.
- Similarly, the combinational logic in a full-scan circuit has two types of outputs: primary outputs (POs) and pseudo primary outputs (PPOs).

Full-Scan Test Operation

- Two test vectors, V1 and V2, are applied to the circuit.
- SE is first set to 1 to operate the circuit in shift mode, and three clock pulses are applied to load the PPI portion of V1 into the scan cells.
- The capture operation is conducted by applying one clock pulse to capture the test response of the combinational logic into the scan cells.

Full-Scan Test Operation

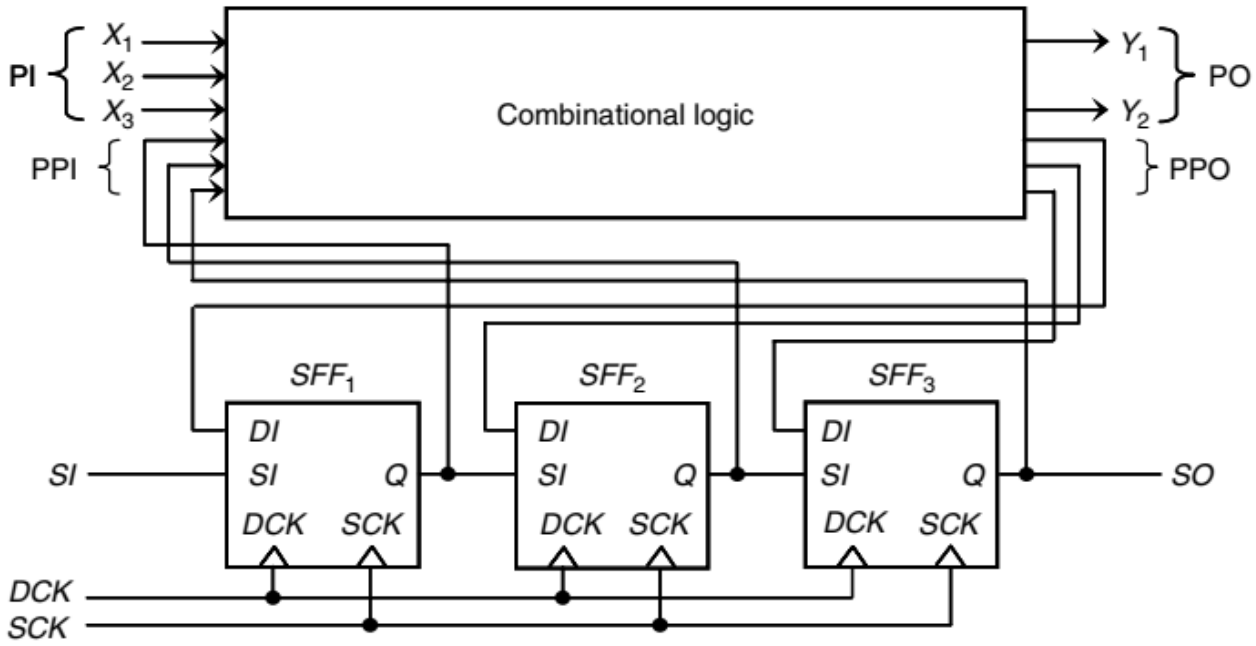
TABLE 2.6 ■ Circuit Operation Type and Scan Cell Mode

Circuit Operation Type	**Scan Cell Mode**	***TM***	***SE***
Normal	Normal	0	0
Shift operation	Shift	1	1
Capture operation	Capture	1	0

Clocked Full-Scan Design

- Figure 2.15 shows a clocked full-scan circuit implementation of the circuit given in Figure 2.13.
- This clocked full-scan circuit is tested using shift and capture operations, similar to a muxed-D full-scan circuit.
- In the clocked full-scan circuit, shift and capture operations are distinguished by properly applying the two independent clocks SCK and DCK during shift mode and capture mode, respectively.

Clocked Full-Scan Design



■ **FIGURE 2.15**

Clocked full-scan circuit.

LSSD Full-Scan Design

- **LSSD Full-Scan Design**

  (See Section 2.5.1.3 in VLSI Test Principles and Architectures)